sysMonitorData MQTT/TLS/JSON

Interface Design Description

Service ID: *"sysMonitorData"*

Abstract
This document describes the MQTT protocol with TLS payload security and JSON payload encoding variant of the `sysMonitorData` service of the SysMonitor support system (SysMonitor-Data-MQTT-IDD-5.2.2).

Contents

1 Overview **3**
1.1 Interface registration 3

2 Service Operations **4**
2.1 operation subscribe 4

3 Data Models **5**
3.1 struct StoredObservation 5
3.2 Primitives 5

4 References **6**

5 Revision History **7**
5.1 Amendments 7
5.2 Quality Assurance 7

1 Overview

This document describes the `sysMonitorData` service interface, which provides the stream of all received observations, republished on the local MQTT broker.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	MQTT	5.0
Data encryption	TLS	1.3 [1], 1.2 [2]
Encoding	JSON	RFC 8259 [3]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `sysMonitorData` service interface

This document provides the Interface Design Description IDD to the *sysMonitorData – Service Description* document [4].

1.1 Interface registration

```
{
  "serviceDefinitionName": "sysMonitorData",
  "version": "1.0.0",
  "metadata": {
    "contract": "SysMonitor-Data-MQTT-IDD-5.2.2",
    "schemaVersion": "1.0"
  },
  "interfaces": [
    {
      "templateName": "generic_mqtts",
      "protocol": "ssl",
      "policy": "CERT_AUTH",
      "properties": {
        "accessAddresses": [
          "mqtt-broker"
        ],
        "accessPort": 8883,
        "baseTopic": "arrowhead/TestCloud/sysmonitor/data",
        "operations": [
          "subscribe"
        ]
      }
    }
  ]
}
```

Listing 1: Registered service instance (TestCloud).

2 Service Operations

2.1 SUBSCRIBE `<baseTopic>/<observationType>/<providerSystem>`

Operation: subscribe
Output: StoredObservation

A consumer connects to the MQTT broker with mutual TLS (its system certificate) and subscribes below the base topic, e.g. `arrowhead/TestCloud/sysmonitor/data/echo/+` or `arrowhead/TestCloud/sysmonitor/data/+/+`. SysMonitor publishes every stored observation with QoS 1, not retained, with `publishedTime` set; the broker's access list decides who may subscribe.

```
{
  "schemaVersion": "string",
  "observationId": "DataManager-echo-1792670400000-42",
  "providerSystem": "DataManager",
  "serviceInstance": "DataManager",
  "observationType": "echo",
  "sourceTime": "2026-10-22T12:00:00.000Z",
  "receivedTime": "2026-10-22T12:00:00.120Z",
  "publishedTime": "2026-10-22T12:00:00.300Z",
  "value": {
    "status": "alive"
  },
  "quality": "good",
  "pipeline": {
    "routingKey": "TestCloud.sysmonitor.DataManager.DataManager.echo"
  }
}
```

Listing 2: A published observation.

3 Data Models

3.1 struct StoredObservation

Object Field	Value Type	Description
"schemaVersion"	String	`1.0`.
"observationId"	String	Identifier (publisher's, or a SHA-256 of routing key and content).
"providerSystem"	String	Publishing system.
"serviceInstance"	String	Service instance.
"observationType"	String	Observation type.
"sourceTime"	DateTime	Publisher time, UTC.
"receivedTime"	DateTime	Time SysMonitor received it.
"publishedTime"	DateTime	Time it was forwarded on `sysMonitorData`.
"value"	Object	Observed value.
"unit"	String	Unit.
"quality"	String	Quality.
"pipeline"	Object	Intake details (routing key, exchange).

3.2 Primitives

Type	Description
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[2] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[3] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[4] J. Delsing, “sysMonitorData – Service Description (SysMonitor),” Eclipse Arrowhead, SD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorData-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	